ETAS RTA-BSW v12.7.0

RTA-BSW CERT-C Compliance Statement

www.etas.com

Copyright

The data in this document may not be altered or amended without special notification from ETAS GmbH. ETAS GmbH undertakes no further obligation in relation to this document. The software described in it can only be used if the customer is in possession of a general license agreement or single license. Using and copying is only allowed in concurrence with the specifications stipulated in the contract.

Under no circumstances may any part of this document be copied, reproduced, transmitted, stored in a retrieval system or translated into another language without the express written permission of ETAS GmbH.

© Copyright 2025 ETAS GmbH, Stuttgart

The names and designations used in this document are trademarks or brands belonging to the respective owners.

ETAS RTA-BSW 12.7.0 - RTA-BSW CERT-C Compliance Statement R01 EN – 05.2025

Contents

1 RTA-BSW CERT-C Compliance Statement 4

1.1 Introduction 4

1.2 Supported CERT-C rules 4

2 RTA-BSW Stacks 6

2.1 Common 6

2.2 RTA-BASE 7

2.3 RTA-CAN 7

2.4 RTA-COM 8

2.5 RTA-DIAG 9

2.6 RTA-ETH 9

2.7 RTA-ETHSEC 12

2.8 RTA-FRAY 12

2.9 RTA-HWD(Can) 13

2.10 RTA-HWD(Eth) 13

2.11 RTA-HWD(Fray) 14

2.12 RTA-HWD(Lin) 14

2.13 RTA-J1939 14

2.14 RTA-LIN 14

2.15 RTA-MEM 15

2.16 RTA-RipsPlugins 16

2.17 RTA-SAFE 16

2.18 RTA-SEC 16

2.19 RTA-SWCL 19

2.20 RTA-XCP 19

3 Contacting ETAS 20

3.1 Technical Support 20

3.2 ETAS HQ 20

3.3 ETAS Local Sales and Support Offices 20

1 RTA-BSW CERT-C Compliance Statement

1.1 Introduction

RTA-BSW consists of a large number of modules grouped into stacks, which are documented in the RTA-BSW reference guides (see `*_Reference_Guide_EN.pdf` in the RTA-BSW documentation directory). An ECU project which uses RTA-BSW modules will configure and generate an ECU specific version of the module using ETAS ISOLAR-AB.

This CERT-C compliance statement follows the reference guide structure, with individual sections for each stack containing a subsection for each module within the stack. Each module subsection lists the CERT-C L1 rules which may be violated by the generated module.

The lists provided are an overview of the violations that should be expected.

When integrating RTA-BSW into a CERT-C compliant project, the integrator should review the generated code for commented rule violations, using their own scans, and check that each violation is acceptable for the project.

1.2 Supported CERT-C rules

The rules are taken from https://wiki.sei.cmu.edu/confluence/display/c/G-G.+Risk+Assessments dated Mar 15, 2022. All L1 rules are covered, and these are printed in the table below

Table 1.1.SEI CERT-C L1 Rules

Rule	Description
ARR38-C	Guarantee that library functions do not form invalid pointers
ENV32-C	All exit handlers must return normally
ENV33-C	Do not call system
ERR33-C	Detect and handle standard library errors
EXP33-C	Do not read uninitialized memory
EXP34-C	Do not dereference null pointers
FIO30-C	Exclude user input from format strings
FIO34-C	Distinguish between characters read from a file and EOF or WEOF
FIO37-C	Do not assume that fgets() or fgetws( ) returns a nonempty string when successful
MEM30-C	Do not access freed memory
MEM34-C	Only free memory allocated dynamically
MSC32-C	Properly seed pseudorandom number generators
MSC33-C	Do not pass invalid data to the asctime() function
MSC41-C	Never hard code sensitive information
POS30-C	Use the readlink() function properly
POS35-C	Avoid race conditions while checking for the existence of a symbolic link
POS36-C	Observe correct revocation order while relinquishing privileges

POS37-C	Ensure that privilege relinquishment is successful
POS39-C	Use the correct byte ordering when transferring data between systems
POS47-C	Do not use threads that can be canceled asynchronously
POS54-C	Detect and handle POSIX library errors
SIG30-C	Call only asynchronous-safe functions within signal handlers
STR31-C	Call only asynchronous-safe functions within signal handlers
STR32-C	Do not pass a non-null-terminated character sequence to a library function that expects a string
STR38-C	Do not confuse narrow and wide strings and functions
INT30-C	Ensure that unsigned integer operations do not wrap

2 RTA-BSW Stacks

2.1 Common

2.1.1 Bfx

Compliant to CERT-C L1 ruleset

2.1.2 Compiler

Compliant to CERT-C L1 ruleset

2.1.3 ComStack

Compliant to CERT-C L1 ruleset

2.1.4 Crc

Compliant to CERT-C L1 ruleset

2.1.5 EcuC

Compliant to CERT-C L1 ruleset

2.1.6 Efx

Compliant to CERT-C L1 ruleset

2.1.7 Ifl

Compliant to CERT-C L1 ruleset

2.1.8 Ifx

Compliant to CERT-C L1 ruleset

2.1.9 Mfl

Compliant to CERT-C L1 ruleset

2.1.10 Mfx

Compliant to CERT-C L1 ruleset

2.1.11 Platform

Compliant to CERT-C L1 ruleset

2.1.12 rba_ArxmlGen

Compliant to CERT-C L1 ruleset

2.1.13 rba_BswSrv

Compliant to CERT-C L1 ruleset

2.1.14 rba_DiagLib

Compliant to CERT-C L1 ruleset

2.1.15 Standard

Compliant to CERT-C L1 ruleset

2.2 RTA-BASE

2.2.1 BswM (BSW Mode Manager)

Compliant to CERT-C L1 ruleset

2.2.2 BswM_PB_Variant_Only

Compliant to CERT-C L1 ruleset

2.2.3 BswM_PreCompile_and_PB_Variant

Compliant to CERT-C L1 ruleset

2.2.4 Det (Default Error Tracer)

Compliant to CERT-C L1 ruleset

2.2.5 EcuM (ECU State Manager)

Compliant to CERT-C L1 ruleset

2.2.6 StbM (Synchronized Time-Base Manager)

Compliant to CERT-C L1 ruleset

2.2.7 Tm

Compliant to CERT-C L1 ruleset

2.3 RTA-CAN

2.3.1 CanIf (CAN Interface)

Compliant to CERT-C L1 ruleset

2.3.2 CanNm (CAN Network Management)

Compliant to CERT-C L1 ruleset

2.3.3 CanNm_PB_Variant_Only

Compliant to CERT-C L1 ruleset

2.3.4 CanNm_PreCompile_and_PB_Variant

Compliant to CERT-C L1 ruleset

2.3.5 CanSM (CAN State Manager)

Compliant to CERT-C L1 ruleset

2.3.6 CanTp (CAN Transport Layer)

Compliant to CERT-C L1 ruleset

2.3.7 CanTp_PBL

Compliant to CERT-C L1 ruleset

2.3.8 CanTp_PBS

Compliant to CERT-C L1 ruleset

2.3.9 CanTp_PC

Compliant to CERT-C L1 ruleset

2.3.10 CanTSyn (Time Synchronization over CAN)

Compliant to CERT-C L1 ruleset

2.4 RTA-COM

2.4.1 Com (Communication)

Compliant to CERT-C L1 ruleset

2.4.2 ComM (Communication Manager)

Compliant to CERT-C L1 ruleset

2.4.3 ComXf (ComBased Transformer)

Compliant to CERT-C L1 ruleset

2.4.4 IpduM (I-Pdu Multiplexer)

Compliant to CERT-C L1 ruleset

2.4.5 LdCom (Efficient COM for Large Data)

Compliant to CERT-C L1 ruleset

2.4.6 Mirror

Compliant to CERT-C L1 ruleset

2.4.7 Nm (Network Manageemnt Interface)

Compliant to CERT-C L1 ruleset

2.4.8 PduR (PDU Router)

Compliant to CERT-C L1 ruleset

2.4.9 rba_ComStackCdd (Rba Comstack Complex Device Driver)

Compliant to CERT-C L1 ruleset

2.4.10 rba_XCoreCdd

Compliant to CERT-C L1 ruleset

2.4.11 rba_Xfrm

Compliant to CERT-C L1 ruleset

2.5 RTA-DIAG

2.5.1 Dcm (Diagnostic Communication Manager)

Compliant to CERT-C L1 ruleset

2.5.2 Dem (Diagnostic Event Manager)

Compliant to CERT-C L1 ruleset

2.5.3 Dlt (Diagnostic Log and Trace)

Compliant to CERT-C L1 ruleset

2.5.4 FiM (Function Inhibition Manager)

Compliant to CERT-C L1 ruleset

2.5.5 FiM_PBL

Compliant to CERT-C L1 ruleset

2.5.6 FiM_PBS

Compliant to CERT-C L1 ruleset

2.5.7 FiM_PC

Compliant to CERT-C L1 ruleset

2.5.8 rba_DemObdBasic

Compliant to CERT-C L1 ruleset

2.6 RTA-ETH

2.6.1 DoIP (Diagnostic Over IP)

Compliant to CERT-C L1 ruleset

2.6.2 DoIP_PBL

Compliant to CERT-C L1 ruleset

2.6.3 DoIP_PBS

Compliant to CERT-C L1 ruleset

2.6.4 DoIP_PC

Compliant to CERT-C L1 ruleset

2.6.5 EthIf (Ethernet Interface)

Compliant to CERT-C L1 ruleset

2.6.6 EthSM (Ethernet State Manager)

Compliant to CERT-C L1 ruleset

2.6.7 EthTSyn (Time Synchronization over Ethernet)

Compliant to CERT-C L1 ruleset

2.6.8 rba_EthArp

Compliant to CERT-C L1 ruleset

2.6.9 rba_EthAutoIp

Compliant to CERT-C L1 ruleset

2.6.10 rba_EthDHCP

Compliant to CERT-C L1 ruleset

2.6.11 rba_EthDHCPv6

- RULE EXP34-C: The messages REQUEST, RENEW, RELEASE and DECLINE are always build and sent after a DHCPv6 server has been selected. Therefore, adrClientData_pcst→adrSelectedServer_pst is always different from NULL.
- RULE EXP34-C: The messages REQUEST, RENEW, RELEASE and DECLINE are always built and sent after a DHCPv6 server has been selected. Therefore, adrClientData_pcst→adrSelectedServer_pst is never NULL referenced.
- RULE EXP34-C: The option “Server ID” is added only to messages REQUEST, RENEW, RELEASE and DECLINE which are always built and sent after a DHCPv6 server has been selected. Therefore, adrClientData_pcst→adrSelectedServer_pst is never NULL referenced.

2.6.12 rba_EthICMP

Compliant to CERT-C L1 ruleset

2.6.13 rba_EthIcmpV6

Compliant to CERT-C L1 ruleset

2.6.14 rba_EthIPv4

- RULE EXP34-C: The ReassemblyBufDescriptor_pst pointer is a valid pointer, since this function rba_EthIPv4_FindInitialHolePos is called by rba_EthIPv4_MergeFragIntoReassemblBuf() only if a valid reassembly buffer exists.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid pointer, because the if loop is entered only when the check of ReassemblyBufCn-t_cu8 is not equal to lLoopIdx_u8 passes, indicating the existence of a valid reassembly buffer.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid pointer, since the else loop is entered only if a valid reassembly buffer is allocated.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid pointer, since this function rba_EthIPv4_MergeFragIntoReassemblBuf is called by rba_EthIPv4_ReassemblyHandler() only if a valid reassembly buffer exists.

2.6.15 rba_EthIPv6

- RULE EXP34-C: The ReassemblyBufDescriptor_cpcst pointer is a valid pointer, since this function rba_EthIPv6_FindInitialHolePos is called by rba_EthIPv6_MergeFragIntoReassemblBuf() only if a valid reassembly buffer exists.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid pointer, since the else loop is entered only if a valid reassembly buffer is allocated.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid pointer, since this function rba_EthIPv6_MergeFragIntoReassemblBuf is called by rba_EthIPv6_ReassemblyHandler() only if a valid reassembly buffer exists.
- RULE EXP34-C: The lReassemblyBufDescriptor_pst pointer is a valid point-er,because the if loop is entered only when the check of ReassemblyBufCn-t_cu8 is not equal to lcheck_buffer_config_u8 passes, indicating the exis-tence of a valid reassembly buffer.

2.6.16 rba_EthNdp

Compliant to CERT-C L1 ruleset

2.6.17 rba_EthTcp

Compliant to CERT-C L1 ruleset

2.6.18 rba_EthUdp

Compliant to CERT-C L1 ruleset

2.6.19 Sd (Service Discovery)

Compliant to CERT-C L1 ruleset

2.6.20 SoAd (Socket Adapter)

Compliant to CERT-C L1 ruleset

2.6.21 SomeIpTp

Compliant to CERT-C L1 ruleset

2.6.22 SomeIpXf (SOME/IP Transformer)

Compliant to CERT-C L1 ruleset

2.6.23 Tcplp (TCP/IP Stack)

Compliant to CERT-C L1 ruleset

2.6.24 UdpNm (UDP Network Management)

Compliant to CERT-C L1 ruleset

2.7 RTA-ETHSEC

2.7.1 rba_EthAh

Compliant to CERT-C L1 ruleset

2.7.2 rba_EthIke2

Compliant to CERT-C L1 ruleset

2.7.3 rba_EthIPsec

- RULE EXP34-C: adrLocalEnd_pcst will not be dereferenced when it is a NULL pointer (See MISRA comment for further information).
- RULE EXP34-C: adrLocalStart_pcst will not be dereferenced when it is a NULL pointer (See MISRA comment for further information).

2.7.4 rba_EthTls

Compliant to CERT-C L1 ruleset

2.8 RTA-FRAY

2.8.1 FrIf (FlexRay Interface)

Compliant to CERT-C L1 ruleset

2.8.2 FrNm (FlexRay Network Management)

Compliant to CERT-C L1 ruleset

2.8.3 FrNm_PB_Variant_Only

Compliant to CERT-C L1 ruleset

2.8.4 FrNm_PreCompile_and_PB_Variant

Compliant to CERT-C L1 ruleset

2.8.5 FrSM (FlexRay State Manager)

Compliant to CERT-C L1 ruleset

2.8.6 FrTp (FlexRay Transport Protocol)

- RULE STR31-C ARR38-C : FrTp_Prv_InitAllQueues Function is called from FrTp_Init and check for configPtr as NULL_PTR is already available in FrTp_Init So can never be Dereference of an invalid pointer value. In post build varients configptr will be passed as a argument to init function.

2.8.7 FrTp_PB_Loadable

Compliant to CERT-C L1 ruleset

2.8.8 FrTp_PB_Selectable

Compliant to CERT-C L1 ruleset

2.8.9 FrTp_PreCompile

Compliant to CERT-C L1 ruleset

2.9 RTA-HWD(Can)

2.9.1 CanTrcv (Can Transceiver)

Compliant to CERT-C L1 ruleset

2.10 RTA-HWD(Eth)

2.10.1 EthTrcv (Ethernet Tranceiver)

Compliant to CERT-C L1 ruleset

2.10.2 EthTrcv_11_Common

Compliant to CERT-C L1 ruleset

2.10.3 EthTrcv_11_StdAutomPhy

Compliant to CERT-C L1 ruleset

2.11 RTA-HWD(Fray)

2.11.1 FrTrcv (FlexRay Transceiver Driver)

Compliant to CERT-C L1 ruleset

2.12 RTA-HWD(Lin)

2.12.1 LinTrcv (Lin Tranceiver)

Compliant to CERT-C L1 ruleset

2.13 RTA-J1939

2.13.1 J1939Dcm (J1939 Diagnostic Communication Manager)

Compliant to CERT-C L1 ruleset

2.13.2 J1939Nm (J1939 Network Management)

Compliant to CERT-C L1 ruleset

2.13.3 J1939Rm (J1939 Request Manager)

Compliant to CERT-C L1 ruleset

2.13.4 J1939Tp (J1939 Transport Protocol)

Compliant to CERT-C L1 ruleset

2.13.5 J1939Tp_PB_Loadable

Compliant to CERT-C L1 ruleset

2.13.6 J1939Tp_PB_Selectable

Compliant to CERT-C L1 ruleset

2.13.7 J1939Tp_PreCompile

Compliant to CERT-C L1 ruleset

2.14 RTA-LIN

2.14.1 LinIf (Lin Interface and Lin Transport Protocol)

Compliant to CERT-C L1 ruleset

2.14.2 LinSM (Lin State Machine)

Compliant to CERT-C L1 ruleset

2.15 RTA-MEM

2.15.1 Ea (EEPROM Abstraction)

Compliant to CERT-C L1 ruleset

2.15.2 Fee (Flash EEPROM Emulation)

Compliant to CERT-C L1 ruleset

2.15.3 MemAcc

Compliant to CERT-C L1 ruleset

2.15.4 MemIf (Memory Abstraction Interface)

Compliant to CERT-C L1 ruleset

2.15.5 NvM (NVRAM Manager)

- RULE ARR38-C,STR31-C: Index is always in bounds, Pointer value is always valid per configuration.
- RULE EXP33-C: All relevant elements of structure jobCryptoData_st are initialized before using it.
- RULE EXP33-C: The sub-structure JobData_st is initialized completely in NvM_Prv_Job_PrepareData just before using it.

2.15.6 rba_EaFs2

Compliant to CERT-C L1 ruleset

2.15.7 rba_FeeFs1x

Compliant to CERT-C L1 ruleset

2.15.8 rba_FeeFs2

Compliant to CERT-C L1 ruleset

2.15.9 rba_FeeFs3

Compliant to CERT-C L1 ruleset

2.15.10 rba_MemLib

Compliant to CERT-C L1 ruleset

2.15.11 rba_MemWrprMemAcc

Compliant to CERT-C L1 ruleset

2.16 RTA-RipsPlugins

2.16.1 Rte_Rips (RTE Implementation Plugin Support)

Compliant to CERT-C L1 ruleset

2.16.2 Rte_Rips_CSSafety

Compliant to CERT-C L1 ruleset

2.16.3 Rte_Rips_CSXfrm (RTE Implementation Plugin Support for C/S Communication with Transformers)

Compliant to CERT-C L1 ruleset

2.16.4 Rte_Rips_Sig2Srv

Compliant to CERT-C L1 ruleset

2.16.5 Rte_Rips_SRCom

- RULE STR38-C: The structure name is different but the elements are the same.

2.17 RTA-SAFE

2.17.1 E2E (End to End Library)

Compliant to CERT-C L1 ruleset

2.17.2 E2EXf (End to End Library Transformer)

Compliant to CERT-C L1 ruleset

2.17.3 WdgIf (Watchdog Interface)

Compliant to CERT-C L1 ruleset

2.17.4 WdgM (Watchdog Manager)

Compliant to CERT-C L1 ruleset

2.18 RTA-SEC

2.18.1 CryIf (Crypto Interface)

Compliant to CERT-C L1 ruleset

2.18.2 Crypto

Compliant to CERT-C L1 ruleset

2.18.3 Csm (Crypto Service Manager)

Compliant to CERT-C L1 ruleset

2.18.4 IdsM

Compliant to CERT-C L1 ruleset

2.18.5 KeyM

Compliant to CERT-C L1 ruleset

2.18.6 rba_CryptoAuCSC

Compliant to CERT-C L1 ruleset

2.18.7 rba_CryptoAuHSM3

Compliant to CERT-C L1 ruleset

2.18.8 rba_CryptoCCL

Compliant to CERT-C L1 ruleset

2.18.9 rba_CryptoHSM

- RULE ARR38-C, STR31-C: objectId_u32 is guaranteed by generators to be less than rba_CryptoHSM_Prv_DriverObjectContext_ast array length
- RULE ARR38-C, STR31-C: objectId_u32 is guaranteed by rba_CryptoHSM_ProcessJob's check to be less than the rba_CryptoHSM_Prv_-DriverObjectContext_ast array length
- RULE ARR38-C, STR31-C: objectId_u32 is guaranteed by the checks of both rba_CryptoHSM_ProcessJob and rba_CryptoHSM_CancelJob to be less than rba_CryptoHSM_Prv_DriverObjectContext_ast array length
- RULE ARR38-C, STR31-C: objectId_u32 is guaranteed by the checks of both rba_CryptoHSM_ProcessJob and rba_CryptoHSM_CancelJob to be less than the rba_CryptoHSM_Prv_DriverObjectContext_ast array length
- RULE EXP34-C: the queue pointer is always valid because the caller function derives it from an array where at least one queue is assigned to all driverObjects
- RULE STR31-C, ARR38-C: key_pcau8 size is set based on the maximum keyelement size configured. This code will not be executed if there are no keyelements with this size configured.
- RULE STR31-C: key_pcau8 size is set based on the maximum keyelement size configured. This code will not be executed if there are no keyelements with this size configured.
- RULE STR31-C: objectId_u32 is guaranteed by rba_CryptoHSM_CancelJob's check to be less than the rba_CryptoHSM_Prv_DriverObjectContext_ast array length

- RULE STR31-C: objectId_u32 is guaranteed by the checks of both rba_CryptoHSM_ProcessJob and rba_CryptoHSM_CancelJob check to be less than the rba_CryptoHSM_Prv_DriverObjectContext_ast array length
- RULE STR31: objectId is guaranteed by rba_CryptoHSM_CancelJob's check to be less than the rba_CryptoHSM_Prv_DriverObjectContext_ast array length

2.18.10 SecOC

- RULE ARR38-C, STR31-C: partitionIndex_u32 is guaranteed by generators to be less than the SecOC_Prv_RxPdu_PartitionContextBuffer_auo array length
- RULE EXP34-C: Input parameters are declared as (void) for generic implementation and it is ensured in the code generation that cryptographicPduBufferIn_pu8 is always available and not a NULL_PTR
- RULE EXP34-C: It is ensured in the code generation that cryptographicPduBufferPosition_uo is always available and not a NULL_PTR
- RULE EXP34-C: It is ensured in the code generation that tpBufferPosition_puo is always available and not a NULL_PTR
- RULE EXP34-C: It is ensured that cryptographicPduState_ven is not a NULL_PTR because of a FIFO implementation and checks in the calling function
- RULE EXP34-C: It is ensured that txAuthenticPduCtx_pst is valid, if txSecuredPduCtx_pst is not a NULL_PTR
- RULE INT30-C: Calculated position cryptographicPduBufOut = posnAuthenticPdu_u8 + authenticPduLength_uo is less than single buffer size is done with configuration values checked with validators
- RULE INT30-C: Calculation is validated in config build
- RULE INT30-C: Calculation validated in config build
- RULE INT30-C: Length calc is validated in config build
- RULE INT30-C: Message link is validated in config build
- RULE INT30-C: Pdu length and calc is validated in config build
- RULE INT30-C: Pdu length is validated in config build
- RULE INT30-C: Pdu length validated in config build
- RULE INT30-C: Pdu length validated in config build, plus runtime check TpSduLength is less than pdu length
- RULE INT30-C: Secured area offset and length are validated in config build
- RULE INT30-C: authDataBufferLen calc validated in config build
- RULE INT30-C: payLen_u32 calc validated in config build
- RULE INT30-C: pduLen size validated in config build
- RULE INT30-C: secOCAuthPduHeaderLength size 0 to 4 Bytes

2.19 RTA-SWCL

2.19.1 rba_CCL

Compliant to CERT-C L1 ruleset

2.20 RTA-XCP

2.20.1 Xcp

Compliant to CERT-C L1 ruleset

3 Contacting ETAS

3.1 Technical Support

Technical support is available to all users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office (see below).

The best way to get technical support is by email. Any problems or questions about the use of the product should be sent to:

rta.hotline@etas.com

If you prefer to discuss your problem with the technical support team, you call the support hotline on:

+44 (0)1904 562624.

The hotline is available during normal office hours (0900-1730 GMT/BST).

3.2 ETAS HQ

ETAS GmbH Borsigstrasse 24 70469 Stuttgart Germany	Phone:	+49 711 3423-0
	Fax:	+49 711 3423-2106
	WWW:	www.etas.com

3.3 ETAS Local Sales and Support Offices

Contact details for your local sales office and local technical support team (where available) can be found on the ETAS web site:

ETAS subsidiaries	www.etas.com/en/contact.php
ETAS technical support	www.etas.com/en/hotlines.php